

Terms of Reference (TOR)

Logistics Management

1. Introduction

Bamburi Cement is a leading cement and concrete producer in Eastern Africa with Key manufacturing operations in the Coastal and Eastern regions.

Coastal Region region comprises of

1. Bamburi Cement [Integrated](#) manufacturing plant
2. BSP concrete & RMX Plants
3. Mbaraki terminal
4. Matuga - Prospect for clinker plant

Eastern/Nairobi Region Comprises of

1. Nairobi Grinding plant - Grinding Station
2. Marimbeti - Readymix/concrete plant
3. Kitui Road - Readymix Plant

Following the divestment, the business would like to manage the inbound and outbound logistics on a different application platform to ensure sustainable [material movement](#).

This TOR outlines the requirements and deliverables for migrating, updating and enhancing inbound & outbound processes.

2. Project Overview

Dispatch logistics management aims to improve the level of customer service for business by ensuring highly transparent operations and sustainable material movements.

Considerations to put in place for dispatch logistics management:

- **Order Management and weighbridge system:** The applications should be able to integrate with SAP S4/Hanna ERP. This will ensure transparency and End-to-end order management [for both inbound and outbound materials](#).
- **Modernizing and Improving Functionality:** Adding new features and/or optimizing existing ones to better serve users.

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3. Scope of Work

The scope of this project covers the following tasks:

1. Assessment and Migration of dispatch and logistics application (HODIM):
 - Conduct a comprehensive review of the existing application ([interest to propose a better application](#))
 - Integration of the weighbridges with the dispatching system
 - End- to- End order management
 - Electronic Proof of delivery
2. Site Weighbridges
 - Weighbridges by site

SITE WEIGHBRIDGES AND DIRECTION CAPABILITY				
#	Site	Weighbride role	Indicator Model	Direction capability
1	Nairobi Grinding Station	Cement Weighbridge in	Data-D2008	One way
2	Nairobi Grinding Station	Cement Weighbridge out	Baykon-BX1	Two Way
3	Nairobi Grinding Station	Cement Weighbridge out	Baykon-E1205	Two Way
4	Nairobi Grinding Station	Raw material Weighbridge	Data-D2008	Two Way
5	Nairobi Grinding Station	Raw material Weighbridge	Data-D2008	Two Way
6	Nairobi Grinding Station	Bulk Loading - Silo1	Baykon-BX1	One way
7	Nairobi Grinding Station	Bulk Loading - Silo2	Baykon-BX22S	One way
8	Nairobi Grinding Station	Bulk Loading - Silo3 & 4	1X-KL3101-D3/CQ0002	One way
9	Mombasa Plant	Cement Weighbridge in	DiniArgeo	Two Way
10	Mombasa Plant	Cement Weighbridge out	DiniArgeo	Two Way
11	Mombasa Plant	Raw material Weighbridge	Avery Tronix-E1105	Two Way
12	Mombasa Plant	Raw material Weighbridge	DiniArgeo	Two Way
13	Mombasa Plant	Clinker Loading	MOORANGE-K3000	One way
14	Mombasa Plant	Silo 7 & 8 weighbridge	Avery Tronix-E1205	One way
15	Mombasa Plant	Silo 9 weighbridge	DiniArgeo	One way
16	Mombasa Plant	BSP-Blocks Plant	Data-D2008	Two Way
17	Kitui Road	Cement Weighbridge	Baykon-BX30	One way
18	Kitui Road	Silo	TBC	One way
19	Mbaraki	Cement Weighbridge	DiniArgeo	One way
20	Mbaraki	Rail/Bulk Weighbridge	Avery Tronix-E1205	One way

21	Marimbeti	Raw material Weighbridge	Avery Tronix-E1205	One way
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○ Weighbridges Per site by Direction Capability

Site	Direction capability		
	One way	Two Way	Grand Total
Kitui Road	2		2
Marimbeti	1		1
Mbaraki	2		2
Mombasa Plant	3	5	8
Nairobi Grinding Station	4	4	8
Grand Total	11	9	21

3. Automation Nairobi Grinding Plant Raw Material Currently installed with Automation devices which are physically working

Exit Weighbridge configuration			
No.	PLC Connections (Siemens Logo 8)	Input/Output Coil	Test Status (Physical)
1	Traffic light Exit from plant B- Green	Q3	OK
2	Traffic light Exit from plant B - Red	Q4	OK
3	Traffic light Entrance from plant B - Green	Q1	OK
4	Traffic light Entrance from plant B - Red	Q2	OK
5	Traffic Light Exit to the Plant A - Green	Q7	OK
6	Traffic Light Exit to the Plant A - Red	Q8	OK
7	Traffic Light Entrance to the Plant A - Green	Q5	OK
8	Traffic Light Entrance to the Plant A - Red	Q6	OK
9	Boom Barrier B	Q9	OK
10	Boom Barrier A	Q10	OK
11	Vehicle Detector B	i1	OK
12	Vehicle Detector A	i2	OK
13	Positional Sensor A	i3	OK
14	Positional Sensor B	i4	OK
15	Positional Sensor C	i5	OK

16	Camera A (ANPR - HIKVision iDS-2CD7A46G0-IZHS)		OK
17	Camera B (ANPR - HIKVision iDS-2CD7A46G0-IZHS)		OK

Entry Weighbridge configuration			
No.	PLC Connections (Siemens Logo 8)	Input/Output Coil	Test Status (Physical)
1	Traffic light Exit from plant B- Green	Q3	OK
2	Traffic light Exit from plant B - Red	Q4	OK
3	Traffic light Entrance from plant B - Green	Q1	OK
4	Traffic light Entrance from plant B - Red	Q2	OK
5	Traffic Light Exit to the Plant A - Green	Q7	OK
6	Traffic Light Exit to the Plant A - Red	Q8	OK
7	Traffic Light Entrance to the Plant A - Green	Q5	OK
8	Traffic Light Entrance to the Plant A - Red	Q6	OK
9	Boom Barrier B	Q9	OK
10	Boom Barrier A	Q10	OK
11	Vehicle Detector B	i1	OK
12	Vehicle Detector A	i2	OK
13	Positional Sensor A	i3	OK
14	Positional Sensor B	i4	OK
15	Positional Sensor C	i5	OK
16	Camera A (ANPR - HIKVision iDS-2CD7A46G0-IZHS)		OK
17	Camera B (ANPR - HIKVision iDS-2CD7A46G0-IZHS)		OK

4. Deliverables

The following deliverables are expected of the weighbridge application:

- **Vehicle Control:** A measurement result will not be provided if the weighbridge platform is not clear of any vehicular/pedestrian traffic or obstruction prior to the vehicle entering the weighbridge
- **Zero Control:** A measurement result should not be provided if the weighbridge platform is not correctly on zero immediately prior to the vehicle entering the weighbridge

- **Zero-setting:** Where a weighbridge platform is not correctly set to zero when the platform is clear, there should be a means of re-setting the indication to zero prior to a truck entering the weighbridge.
- **Truck Position:** A measurement should not be provided unless the truck is correctly positioned on the weighbridge platform.
- **Software:** The software used to acquire measurement data should not be conducive to fraudulent modification. (minimal human interaction if not none)
- Offline mode?
- Tolerance setting in the software?
- Weighbridge off calibration detection capability of the software?
- Printed records shall include the following information and comply with the requirement for transaction records
 - Unique transaction identifier
 - Location of the measurement
 - Identity of the controller of the weighbridge
 - Measurement date and data
 - Vehicle identity
 - Gross Weight
 - Tare Weight
 - Net Weight
 - Axle Weight
- **Stored Tare Weight:** Tare weights captured by the operation of the weighbridge should be stored by the control system (ERP) for future reference
- **Two Step Process for inter-company transfers**
 - Outbound Delivery -> Stock-in-Transit -> Inbound Delivery
 - Containerized clinker transfer capturing for the automation process
- **TTAT (Truck Turn Around Time):** In addition to the above features, the system should be able to accurately track and report/produce analytics on the operations cycle and determine the time by trucks within the business premises at different points of interests but not limited to Truck inspection, order assignment, 1st weight, loading time, 2nd weight and exit both for inbound and outbound transactions.
- **Electronic Proof of delivery (ePOD):** Validation of goods receipt by the customer
- Reconciliation report from the Dispatch tool to ERP
- Error feedback for case of Good movement (GRN not done)
- Audit Trail

5. Timeline

Following the divestment, we look forward to have this project completed as per the following key milestones:

- Initial assessment and planning
 - Application deliverable assessment
 - Assess training needs
- Master Data migration
- Training of internal teams
- Phased Roll-out of the new application to all sites
- Migration to the new application
- Initial test run identification of gaps and closure
- Safety Check-list
- Go-LIVE
- Stabilization & Post-launch support.

Process Flows - Appendix

- 1. Inter-Plant Transfer Process Flow**
- 2. Containerized Clinker Process**
- 3. Inbound Process**
- 4. OutBound Process Flow**
- 5. Weighbridge Robotics**

6. Key Stakeholders

- Bamburi Cement's Logistics and IT teams.

7. Evaluation Criteria

Proposals for this project will be evaluated based on:

- Demonstrated understanding of the business diversification process leading to change of current dispatch and logistics applications.
- Proposed approach to addressing identified issues.
- Experience in similar projects and relevant portfolio examples.
- Project timeline and cost-effectiveness.

8. Reporting & Communication

- Bi-Weekly progress meetings with key stakeholders.
- Approval required for design concepts and new functionality before implementation.

Approvers

Request			
Designation	Name	Signature	Date
Logistics Manager - MOM	Mercy Kirui		
Logistics Manager - Nai	Anita Njiru		
Logistics Operations Manager	James KITHUKA		
Project Leader (IT EUSTL - KENYA)	Hillary Musyoki		
Business Solutions Analyst	Samuel OTIENO		
Country IT Manager	Tom ABWAO		
Approvals			
Supply Chain Director	Moses WERE		
Finance Director	Eugene ANTERA		